VAVR vs Zazr Deep Dive

VAVR vs Zazr · Same Option describe in both libraries · Build with Waqar

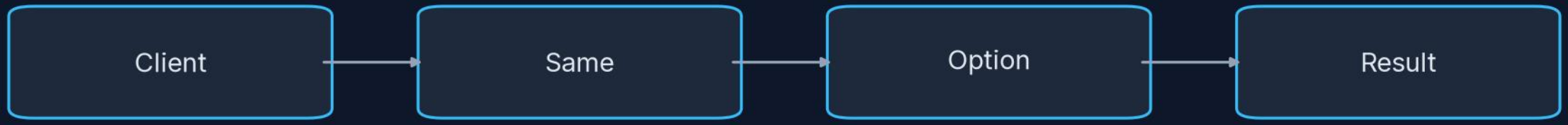
Today's Hook

- Problem: Same Option describe in both libraries
- Pattern: VAVR vs Zazr — earned from the pain, not trivia.
- Runnable main on JDK 25 with captured stdout.
- Naive → solution → failure paths → trade-offs.

The Problem

- Same Option describe in both libraries
- Naive design shows the cost clearly.
- Pattern isolates the force behind a clean boundary.
- Tests cover happy path and injected failures.

Structure



VAVR vs Zazr: Same Option describe in both libraries

Real-World Scenario

- Running example: Same Option describe in both libraries
- Same names in naive code, refactor, and Demo output.
- No shared StreamForge framing unless it fits.
- Product language first; pattern vocabulary second.

Naive Code

```java
package com.buildwithwaqar.patterns.functional.vavrcompare;

import com.buildwithwaqar.patterns.functional.vavrcompare.VavrVsZazr;
/** Problem: migrate Option describe from VAVR to Zazr pattern matching. */
public final class UseCaseDemo {
  public static void main(String[] a) {
    System.out.println("=== VAVR vs Zazr problem ===");
    System.out.println("vavr=" + VavrVsZazr.vavrDescribe(io.vavr.control.Option.some(7)));
    System.out.println("zazr=" + VavrVsZazr.zazrDescribe(dev.zazr.control.Option.some(7)));
  }
}
```

Why Naive Hurts

- Call sites couple to the wrong shape or leak on partial failure.
- Parameter drilling / duplicated envelopes / missing rollback.
- Hard to unit-test without the real dependency.
- New variants require shotgun edits.

Solution

```java
package com.buildwithwaqar.patterns.functional.vavrcompare;

/** Side-by-side Option usage — VAVR (classic) vs Zazr (Java 25 sealed/records). */
public final class VavrVsZazr {
    private VavrVsZazr() {}

    public static String vavrDescribe(io.vavr.control.Option<Integer> opt) {
        return opt.map(n -> "got " + n).getOrElse("missing");
    }

    public static String zazrDescribe(dev.zazr.control.Option<Integer> opt) {
        return switch (opt) {
            case dev.zazr.control.Option.Some(var n) -> "got " + n;
            case dev.zazr.control.Option.None() -> "missing";
        };
    }
}
```

Runnable main

```java
package com.buildwithwaqar.patterns.functional.vavrcompare;

import com.buildwithwaqar.patterns.functional.vavrcompare.VavrVsZazr;
/** Problem: migrate Option describe from VAVR to Zazr pattern matching. */
public final class UseCaseDemo {
  public static void main(String[] a) {
    System.out.println("=== VAVR vs Zazr problem ===");
    System.out.println("vavr=" + VavrVsZazr.vavrDescribe(io.vavr.control.Option.some(7)));
    System.out.println("zazr=" + VavrVsZazr.zazrDescribe(dev.zazr.control.Option.some(7)));
  }
}
```

Walkthrough Output

```
=== VAVR vs Zazr problem ===
vavr=got 7
zazr=got 7
```

Step-by-Step Trace

- Each stdout line maps to a stage above.
- Re-run the main class locally on JDK 25.
- Failure-path lines prove compensations / rejects / missing files.
- Pause and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction
// records + sealed + switch
// inject Env / collaborators at the edge
// virtual threads for blocking fan-out
```

Trade-offs

- Clarity vs ceremony — refuse five types for two lines.
- Teaching demos omit production metrics/idempotency keys.
- Prefer simpler language features when they erase the force.
- Know the twin pattern often confused with VAVR vs Zazr.

Related Patterns

- Contrast VAVR vs Zazr with its neighbor pattern.
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Problem → naive → VAVR vs Zazr → Demo stdout.
- Failure paths are part of the lesson.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run the main.